

Construction Sites and Parking Tickets in Toronto*

Construction is not only paid by the government but also by the people. In this report, we show how construction sites affect the number of parking tickets issued in Toronto.

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Construction sites take up curb space in a city where parking is already scarce. We investigated 32 million Toronto parking tickets from 2010 to 2025 and compared blocks where construction started with similar nearby blocks without construction. After construction started, construction blocks received 7.8 percent more tickets than expected, and the gap grew from 4.0 percent in the first six months to 14.1 percent in the second year. Policy makers that permits that construction could come with parking plans for the surrounding blocks.

1 Introduction

In Toronto, the price of a car is only the lower bound of what it costs to own one. Owners also consider the price for fuel, insurance, maintenance and the loss in the car's value over time (Canadian Automobile Association 2018), and in a large city, parking as well (Shoup 2005). Two of these costs depend heavily on where a person lives and works. Insurers in Ontario price premiums partly by postal code (Wadehra 2024) and the cost of parking depends on how many spaces are available nearby (Shoup 2005) at any time of the day (Figure 1). When curb space becomes scarce, it creates business opportunities. Parking on the street can be cheap but hard to find, so businesses take advantage of the lack of options and charge more for off-street parking. As a result, drivers either spend time circling for a space (Shoup 2006) or park illegally, weighing the fine against the chance of being caught (Becker 1968). Toronto issues about two million parking tickets a year (City of Toronto 2026f), so these choices add up. This paper asks what affects the number of tickets written on a street, and whether the City's own policies have unintended consequences for parking.

*Code and data are available at: <https://github.com/FurkanDanisman/toronto-parking-tickets>.

Research on parking has studied how parking prices influence drivers’ choices (Shoup 2005, 2006), and parking tickets have been used to study how enforcement and social norms affect behaviour (Fisman and Miguel 2007). In a city like Toronto, where construction is constant, less attention has been given to how construction influences drivers’ choices. This project addresses this gap by studying the impact of construction on parking ticket issuance. A demolition or a new building often occupies part of the road or sidewalk, which requires a separate street occupation permit from the City, and any parking spaces taken for construction staging must be identified in the application (City of Toronto 2026g). Construction can therefore remove curb space while bringing trucks and workers who also need to park. Using 32,211,737 parking tickets issued in Toronto between 2010 and 2025 (City of Toronto 2026f) and the City’s building permit records (City of Toronto 2026c, 2026d), we analyze the relationship between construction and ticket issuance using causal inference. We compare a treated group of 5,484 blocks where construction started with a control group of nearby blocks on the same type of street that had no construction at the time and had a similar number of tickets in the year before (Stuart 2010). With this difference-in-differences design (Card and Krueger 1994; Angrist and Pischke 2009), we compare how tickets changed on construction blocks with how they changed on similar blocks over the same months.

The remainder of this paper is structured as follows. Section 2 describes the parking ticket and building permit data, how they are measured, and where tickets are written across the city and the day. Section 3 explains how blocks with construction are compared with similar blocks without it and presents the results. Section 4 discusses what the results mean and their limitations. Section A in the appendix describes how the data were cleaned.

2 Data

2.1 Source and context

All data come from Open Data Toronto, the City of Toronto’s open data portal, launched in 2009 (City of Toronto 2026a), and were downloaded with `opendatatatoronto` (Gelfand 2025).

Parking tickets are published annually by the City’s Revenue Services division, one row per ticket since 2008 (City of Toronto 2026f). Tickets are issued by Toronto Police Service personnel and people it has certified. Licence plates are not published and kept anonymous. We use 2010 to 2019 and 2022 to 2025; 2020 and 2021 are excluded because of the pandemic (Westoll 2020).

Information on construction sites comes from Toronto Building’s active and cleared building permits (City of Toronto 2026c, 2026d). Tickets and sites are located with the City’s address points (City of Toronto 2026b) and street centreline (City of Toronto 2026h). While visualizing on maps, we use the former municipality boundaries (City of Toronto 2026e).

Table 1: Variables recorded for each parking ticket, with an example.

Variable	Description	Example
Ticket number	Ticket identifier with the first digits masked	***65822
Date	Day the ticket was issued	1 January 2025
Time	Time on a 24-hour clock	00:02
Infraction code	Number identifying the rule that was broken	29
Infraction description	Short text version of the rule	Park prohibited time no permit
Fine	Set fine in dollars	\$45
Location	Street address, or a street and a cross street	Near 77 Portland St
Province	Province or state of the licence plate	ON

The analysis uses R (R Core Team 2024) with the `tidyverse` (Wickham et al. 2019), `arrow` (Richardson et al. 2024), `sf` (Pebesma and Bivand 2023), `jsonlite` (Ooms 2014), `fixest` (Bergé 2018), `testthat` (Wickham 2011), `here` (Müller 2020), and `knitr` (Xie 2024). Graphs are made with `ggplot2` (Wickham 2016) and `hexbin` (Carr et al. 2024), and tables with `tinytable` (Arel-Bundock 2024).

2.2 Measurement

A ticket is created when an officer sees a vehicle breaking a parking rule and issues a ticket recording the date, time, rule broken, set fine and location, given either as an address or as a street (City of Toronto 2026f).

A construction site is recorded when the City issues a building permit, with an issue date and, once the permit is closed, a completion date (City of Toronto 2026c, 2026d).

2.3 Parking tickets

Table 1 lists what each ticket records.

Table 2 shows between 1.8 million and 2.8 million tickets each year, with the lowest count in 2022. Between 86 and 88 percent of tickets each year could be matched to an address and assigned to a street block; the rest were mostly written at intersections, which have no street number. The average fine increases from \$40 in 2010 to about \$49 in 2016 to 2019, then to \$68

Table 2: Parking tickets issued each year, the share assigned to a street block, the average fine, and the total value of fines, 2010 to 2019 and 2022 to 2025.

Year	Tickets	Placed on a street block	Average fine	Total fines (\$ millions)
2010	2,721,462	86%	\$40	110.0
2011	2,805,492	86%	\$41	114.2
2012	2,746,154	86%	\$41	111.5
2013	2,351,358	87%	\$41	95.8
2014	2,484,983	87%	\$44	109.0
2015	2,168,493	86%	\$46	99.9
2016	2,254,761	87%	\$49	109.7
2017	2,144,226	87%	\$50	106.2
2018	2,044,288	87%	\$49	100.2
2019	2,213,939	86%	\$49	108.4
2022	1,821,887	86%	\$53	96.3
2023	2,256,360	88%	\$54	122.4
2024	2,147,159	88%	\$68	146.0
2025	2,051,175	88%	\$80	164.7

in 2024 and \$80 in 2025. However, one should consider the inflation before making judgements about the fee.

We split the day into four periods: overnight (00:00 to 05:59), morning (06:00 to 08:59), working hours (09:00 to 16:59) and after work (17:00 to 23:59). Figure 1 shows the average number of tickets per hour in 2025 for each period, and the dashed lines mark the boundaries of the six former municipalities. Overnight, tickets are spread fairly evenly across the city. When people start going to work, tickets begin to concentrate downtown, and this concentration is strongest during working hours. After work, the concentration eases and tickets spread out across the city again. This could mean that there is no particular driving culture in one part of the city, but rather that the concentration of tickets is driven by where people work and where they park during the day.

2.4 Construction sites

We identify construction sites from building permits and keep only permits for new buildings and demolitions, since work inside existing buildings, such as plumbing or mechanical work, is less likely to affect the street. Table 3 shows 12,628 demolitions and 18,716 new buildings

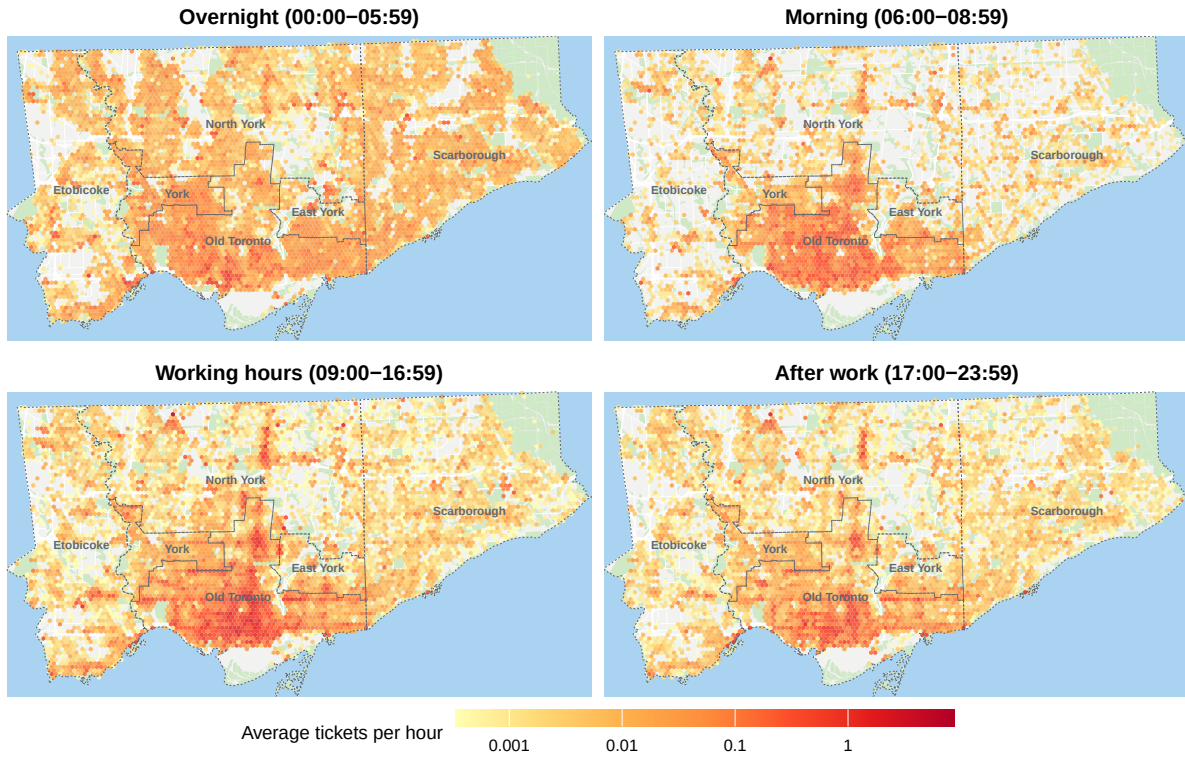


Figure 1: Average parking tickets per hour in 2025 by time of day, in 300-metre hexagons. Dashed lines are the boundaries of the six former municipalities.

Table 3: New buildings and demolitions between 2010 and 2025.

Type of work	Sites	Placed on a street block	Typical months active
Demolition	12,628	92%	6
New Building	18,716	91%	36

active between 2010 and 2025, and about 91 to 92 percent of each could be assigned to a street block.

3 Construction and parking tickets

3.1 Comparing blocks with and without construction

We treat construction as an intervention and measure its influence by forming two groups. The treatment group consists of blocks where construction started, and the control group consists of nearby blocks with no construction that had a similar number of parking tickets in the year before.

Before any inference, we first check the parallel trends assumption by comparing the two groups in the year before construction started. This assumption means that, without construction, both groups would have continued to change in the same way. Then we compare how the two groups changed before and after construction started to estimate the effect of construction on parking tickets.

We estimate the effect of construction with the regression

$$y_{it} = \alpha_i + \gamma_{s(i),t} + \beta D_{it} + \varepsilon_{it}.$$

Each group is followed from 12 months before to 24 months after its construction started. In the model:

- y_{it} is the number of parking tickets on block i in month t .
- α_i is the average number of tickets on block i .
- $\gamma_{s(i),t}$ is the month effect shared by block i 's group $s(i)$ in month t .
- D_{it} is recorded as 1 for a construction block in the months after its construction started, and 0 otherwise.
- β is the number of extra tickets per month on a construction block after construction started.
- ε_{it} is the part of the ticket count the model does not explain.

The model above gives one average effect for all 24 months after construction started. To see how the effect changes over time, we also estimate it separately for each six-month period, from 12 months before to 24 months after construction started. Each estimate in Figure 3 is the number of extra tickets per block per month in that period.

3.2 Results

Before construction started, the two groups moved in parallel (Figure 2), and the difference was only 0.07 tickets per block per month, which is not statistically significant ($p = 0.10$). After construction started, construction blocks received 4.0 percent more tickets than expected in the first six months, 8.0 percent more in months 7 to 12, 11.4 percent more in months 13 to 18 and 14.1 percent more in months 19 to 24 (Figure 3). Over the full 24 months, construction blocks received 7.8 percent more tickets than expected, or in other words, 0.44 extra tickets per block per month ($p < 0.001$).

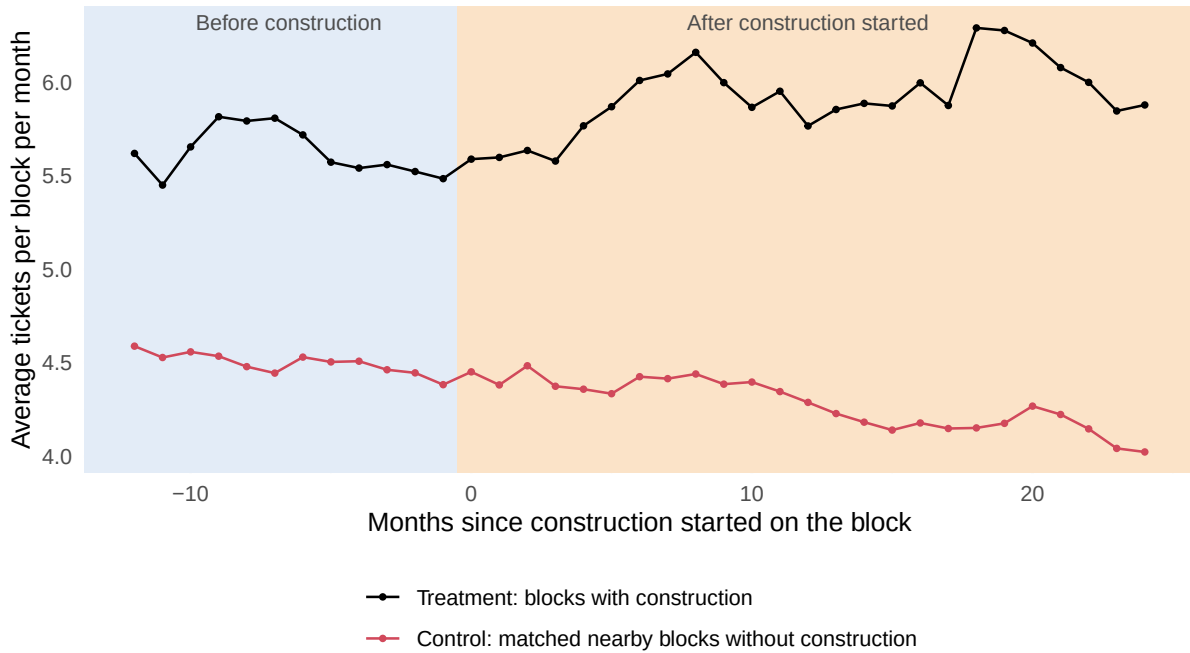


Figure 2: Average parking tickets per block per month on blocks where construction started (treatment) and on their five matched nearby blocks without construction (control), from 12 months before to 24 months after construction started.

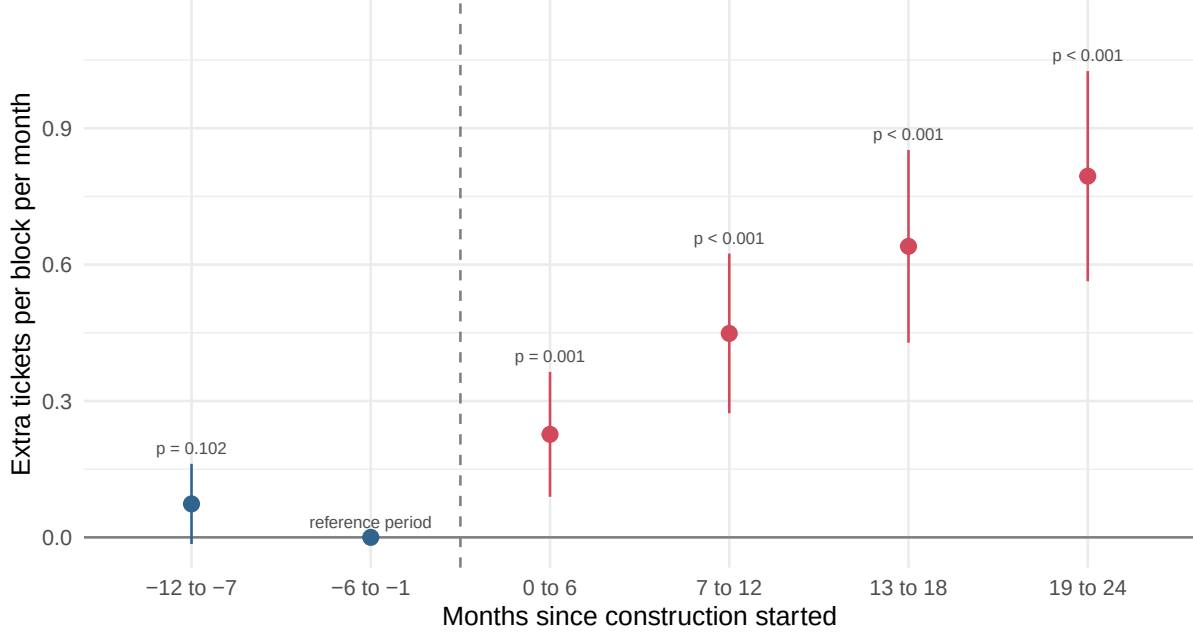


Figure 3: Difference-in-differences estimates of extra parking tickets per block per month on construction blocks, by six-month period, relative to the six months before construction started. Vertical lines are 95 percent confidence intervals.

4 Discussion

In this study we only investigated the areas in Toronto, so the results apply only at the city level; similar results from other cities would strengthen the implications. Beyond generalizability limitation, we are also limited in what we can infer about causes. We do not know whether other changes caused the groups to move in these directions. There may be unmeasured confounders.

Despite these limitations, at the city level we observe that construction significantly affects the number of parking tickets. For the City, these results suggest that permits allowing construction to occupy the street could come with plans for the surrounding blocks, such as temporary parking, loading zones or adjusted enforcement, which would give drivers more options and less reason to park illegally. Also, part of the explanation may be that construction brings temporary parking rules that are unusual or not well known to the public. Either way, in the future studies, the problem can be investigated further, and strategies can be developed to reduce the number of tickets issued in these areas.

Appendix

A Data cleaning

- We downloaded the yearly ticket files directly, because `opendatatoronto` returned only 1.42 million of the 2.75 million tickets for 2012. The 2010 file is UTF-16 encoded and was converted.
- The 2013 file ends on 18 November 2013. We left November and December 2013 out of the construction analysis.
- We standardized street names (for example, “AVENUE” to “AVE”), matched each ticket to the City’s address points, and assigned it to the nearest street block. Tickets written at intersections have no street number and were not assigned.
- We kept building permits for new buildings and demolitions only. We dropped 1,163 permits (about 3 percent) whose completion date is before their issue date. A demolition counts as active for at most six months and a new building for at most three years after its permit was issued.
- For each construction block, we chose five control blocks within 1.5 km but more than 250 m away, on the same type of street, with no construction from 12 months before to 24 months after the start, and with the closest number and trend of tickets in the year before.

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